the Journal of Strength and Conditioning Research™

Football Practices in Hot Environments Impact Subsequent Days' Hydration

Candi D. Ashley,[1] Rebecca M. Lopez,[2] and Amanda J. Tritsch[2]

[1]Department of Educational and Psychological Studies, University of South Florida, Tampa, Florida; and [2]School of Physical Therapy and Rehabilitation Sciences, University of South Florida, Tampa, Florida

Downloaded from http://journals.lww.com/nsca-jscr by Ua4HyonDIxZ3azs6PvPeafPjdRjmqTz2+wOaEOfPRKBWqmu2eTwLZFZCxD413lIzA9vtUjMVkNF8F/e29KG4jsPwh382C35tqtwKdeh21n/tkVAV08GvvOJ+sJCFOPfdY+hIYkKNuETHd/BLe1WCR3X4GJ6GTTKP1jwaHaj28= on 10/11/2023

Abstract

Ashley, CD, Lopez, RM, and Tritsch, AJ. Football practices in hot environments impact subsequent days' hydration. *J Strength Cond Res* XX(X): 000–000, 2023—The impact of proper hydration to prevent exertional heat illness in American football has not been evaluated during high school preseason football practices in a hot environment (wet-bulb globe temperature = 31.3 ± 1.8°C). The purposes of this study were to examine the accuracy of urinary hydration measures to assess body mass (BM) changes and to examine carryover effects of consecutive practices by comparing postpractice with the next prepractice values. Before and after each of 7 outdoor practices, 31 male high school football players (age = 16 ± 1 years, height = 181.2 ± 12.0 cm, BM = 85.7 ± 19.1 kg, body mass index = 20.8 ± 1.8) provided a urine sample and were weighed to assess hydration. Sensitivity and specificity of urine color (Ucol) and urine-specific gravity (USG) to determine BM changes were determined using receiver operating characteristic (ROC) analysis. Paired samples *t*-tests assessed carryover effects between practices. Repeated-measures analysis of variance assessed carryover effects across practices. Significance was set at $p < 0.05$. Sensitivity and specificity for using Ucol or USG to determine BM changes was not significant. For Ucol, there was a carryover effect from practice numbers 2 to 3, 6 to 7 AM, 7 AM to 7 PM ($p < 0.001$ for all), and 10 to 11 ($p = 0.004$); most with less than 24 hours between practices. The %BM loss (%BML) was significantly greater ($p = 0.001$ to 0.024) after 2-a-day practices. Effects of previous days' exercise in the heat, as evidenced by higher Ucol and %BL, are greater after 2-a-day practices, which occurred on later practice days. Athletes must replenish fluids during and between practices to remain euhydrated.

Key Words: heat illness, body mass loss

Introduction

Exertional heat illness (EHI) is common in outdoor exercise in the heat especially in football (9,14,27,29,30,32–34). Kerr et al. (14) evaluated EHIs in myriad sports in 100 high schools over 5 academic years. Of the 300 EHI occurrences, most (72%) occurred in American football with the majority of those (61.6%) occurring during preseason practices (14). Most cases of EHI are preventable with appropriate mitigation strategies, such as proper hydration. Yet there is no research measuring hydration status in high school American football players in hot and humid conditions. The only published studies on the hydration responses of adolescent American football players took place in mild environmental conditions (maximum wet-bulb globe temperature [WBGT] = 15.5–25° C) (26,31). It is currently unknown how more severe environmental conditions may affect the hydration status of young American football players during preseason practices. Furthermore, research has shown that consecutive days of exercise in the heat can increase the risk of EHI (29), but there is no published data on the potential of a negative carryover effect with this younger population. Godek et al. (10) found that over 8 days of preseason practice in collegiate American football players, body mass (BM) was reduced below baseline throughout the 8 days and suggest that this is possibly because of the lack of adequate hydration from consecutive days of practice in the heat.

Address correspondence to Dr. Candi D. Ashley, cashley@usf.edu.

Journal of Strength and Conditioning Research 00(00)/1–7

© 2023 National Strength and Conditioning Association

Typically, hydration status can be evaluated by assessing percent of BM loss (%BML) with changes in %BML during sport and activity and across active days suggested as the preferred field assessment of hydration (4). A %BML of 1% signals the beginning of impaired thermoregulation. A %BML of 2–3% can result in reduced sport performance and impaired thermoregulation (13,15,17). A laboratory study of simulated 2-a-day practices approximately 120 minutes in length (WBGT = 31.9 ° C) in college-aged men resulted in %BML ranging from 0.63 to 1.14% (22). In a recent field study, DeMartini-Nolan et al. (6) reported 1.56% BML in 29 collegiate football players during 8 days of practice in a warm environment (WBGT = 29.6° C). During simulated 2-a-day practices in college football players (WBGT ranged from 23.5 to 26.7° C), approximate %BML in a study by Godek et al. (10) ranged from 1.0 to 1.7% BML.

Urine-specific gravity (USG) is also used to evaluate hydration status (3,5,6,8,26,31). When only a single measurement is available, a USG value greater than 1.020 has been associated with dehydration (3,4,20,22,24). In a study evaluating prepractice USG in 174 NCAA athletes in various sports, 66% of the athletes had a USG of ≥1.020, indicative of hypohydration (28). Field studies that evaluated prepractice and postpractice USG in college-aged men during practices report prepractice values ranging from 1.020 (6) to 1.025 (10) and postpractice values ranging from 1.025 (6) to 1.032 (10). It is noteworthy that baseline USG in the study by Godek et al. (10) was lower than most of the subsequent practices. In examining the USG of high school American football players, prepractice values ranged from 1.020 to 1.025 (26,31) and end-of-day postpractice values ranged from 1.023 to 1.025 (31).

Copyright © 2023 National Strength and Conditioning Association. Unauthorized reproduction of this article is prohibited.

the Journal of Strength and Conditioning Research™

Table 1

Preseason practice schedule with day and environment.*

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Week 1 of preseason practice	Px 1 Indoors	**Px 2** Outdoors	Px 3 Mix	Px 4 Indoors	Px 5 Mix	**Px 6**† Outdoors
Week 2 of preseason practice	**Px 7 AM**† Outdoors **Px 7 PM** Outdoors	**Px 8** Outdoors	Px 9 AM† Mix Px 9 PM Mix	**Px 10** Outdoors	**Px 11**† Outdoors	

*Px = practice. Bold entries indicate outdoor practices.
†All practices occurred in the afternoon except Px 6, Px 7 am, Px 9 am, and Px 11.

The assessment of USG for an athletic team may be time-limiting and requires special equipment, which may not be available or practical in all field settings. Urine color (Ucol) (2) is a noninvasive and inexpensive way to measure hydration status using an 8-color scale ranging from pale yellow to brownish and can been used in the field to determine hydration status (4,6,8,15,31). A Ucol value of greater than or equal to 5 is indictive of significant hypohydration (17). In field studies examining Ucol in American football players, prepractice and postpractice values ranged from 4 to 6 (6,31). McKenzie et al. (16) evaluated the accuracy of Ucol to detect excess BML in college-aged men who cycled in a controlled environment (WBGT = 36° C) for up to 5 hours or to 5% BML. Based on sensitivity and specificity calculations, they determined that Ucol can provide a practical assessment of hydration status.

It should be noted that no single measure of hydration is without limitations. To reduce misclassification of hydration status, multiple assessments, including urinary measures of hydration along with BM changes, will provide a more accurate representation of hydration status (3,15,24). It would seem advantageous to determine appropriate surrogate measures of hydration status that are easily completed in field settings in a hot and humid environment in high school American football players. Therefore, the purposes of this study were: (a) to evaluate the accuracy of Ucol and USG to detect greater than 2% BML and (b) to examine the potential carryover effects of hot practices on the next practice. The results of this study will help elucidate hydration measures to help ensure high school American football players maintain adequate hydration during and between practices.

Methods

Experimental Approach to the Problem

Exertional heat illness, which is common in American football (9,14,27,32–34), can be mitigated with proper hydration (1,4,15,21,24). There are several measures used to examine hydration status, such as Ucol, USG, and %BML (1–6,8,13,15,17,26,31). However, there are no studies examining the hydration status of high school football players under hot and humid conditions. In addition, there is evidence that there is an additive effect of multiple days of exercise in the heat (10,29). As such, this study used a convenience sample of high school football players during preseason practice to determine the utility of using Ucol to and USG to assess hydration and to examine the carryover effect of multiple days of preseason practices.

Subjects

Hydration responses (%BML, Ucol, and USG) of American football players ranging in age from 14 to 18 years from a high school varsity team were subjects (n = 31, mean age, body mass, and height = 15.8 ± 1.0 years, 85.7 ± 19.1 kg, 181.2 ± 12.0 cm, respectively). Athletes had undergone a preparticipation physical before beginning preseason practice. As such, all were deemed fit to participate in preseason practices.

This study was reviewed and approved by the University of South Florida Institutional Review Board. Potential subjects attended an informational meeting where the research study including risks and benefits and the consent form were explained. Subjects and their parent or legal guardian were given an opportunity to ask questions before consenting. Willing subjects provided written assent if younger than 18 years, and their parent or legal guardian provided written informed consent. Willing subjects 18 years of age and older provided written informed consent. Consent and assent were obtained before any data collection. The team's athletic trainer reviewed their medical history forms to ensure that they did not meet any of the exclusionary criteria listed above.

Data were collected across the first 2 weeks of preseason practices for a total of 11 practices. Because of occasional thunderstorms during preseason practices, some practices were conducted indoors, some outdoors, and some a mix of both indoors and outdoors (Table 1). Our data represent outdoor practices only (n = 7 practices) with a mean WBGT of 31.3 ± 1.8° C (Kestral 5400 Heat Stress Tracker). Table 2 provides length of each practice and environmental information for each of the outdoor practices. Figure 1 provides a diagram of data collection procedures.

Procedures

The main outcome measures for this article are changes in body mass and urinary hydration measures (Ucol and USG) between and across practices, and %BML across practices. Wet-bulb globe temperature was recorded throughout practices.

In the locker room, before practice, subjects provided a urine sample for the assessment of Ucol through Armstrong's Ucol chart (2)and USG through refractometry (REF312ATC; General Tools & Instruments, Secaucus, NJ); seminude body mass while wearing shorts only was assessed (Tanita BWB800, Tokyo, Japan).

During football practices, environmental conditions were recorded every 15–20 minutes. Throughout practices, each subject was assigned an individual water bottle from which to drink, and bottles were refilled and measured as needed. Any subject

Table 2

Environmental data for hot practices.*

Practice	Length of practice (min)	Mean WBGT (ºC)
Pr2	75	33.1 ± 0.9
Pr6	120	31.2 ± 3.4
Pr7 AM†	60	29.2 ± 2.0
Pr7 PM	120	32.3 ± 2.0
Pr8	160	31.8 ± 2.2
Pr10	90	32.8 ± 1.1
Pr11†	120	28.4 ± 2.5

*WBGT = wet-bulb globe temperature.
†All practices were in the afternoon except Pr7 am and Pr11, which were conducted in the morning.

Copyright © 2023 National Strength and Conditioning Association. Unauthorized reproduction of this article is prohibited.

Downloaded from http://journals.lww.com/nsca-jscr by Ua4HyonDIxZ3azs6PvPeafPjdRjmqTz2+wOaEOfPRKBWqmu2eTwLZFZCxD413llzA9vtUjMVkNF8F/e29KG4jsPwh382C35tqtwKdeh21nIkVAV08GvvOJ+sJCFOPfdY+hIYkKNuETHd/BLe1WCR3X4GJ6GTTKP1jwaHaj28= on 10/11/2023

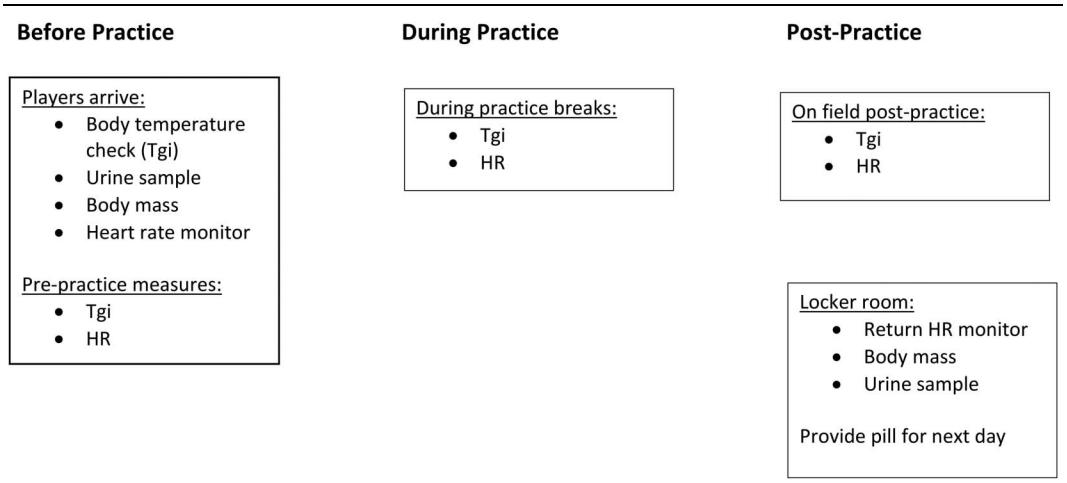

Figure 1. Within-practice study procedures.

who experienced symptoms of heat illness was referred to the school's athletic training staff.

Once in the locker room after practice, subjects towel dried and a postpractice seminude BM was obtained. Along with the prepractice body mass and fluid intake, %BML was calculated. Then, subjects provided a postpractice urine sample for the assessment of Ucol and USG. For the day with 2 football practices (7 AM and 7 PM), the same procedures for data collection were followed for both practices. Subjects were instructed to consume the same food and beverages in the 12 hours before each football practice and between practices for the day with multiple sessions.

Statistical Analyses

The primary analyses for this article are examinations of sensitivity and specificity of using %BML, USG, and Ucol to determine hydration, and potential carryover effects of measures of hydration (Ucol, USG, BM, and %BML) of one outdoor practice on the subsequent practice and across practices. Descriptive data (mean ± *SD*) were calculated for all variables.

The diagnostic utility for using Ucol and USG to determine hypohydrated (%BML ≥2.0) and euhydrated (%BML <2.0) state was evaluated using receiver operating characteristic (ROC) curve analysis. The criterion values for Ucol (hypohydrated ≥5; euhydrated <5) (16,17) and USG (hypohydrated >1.020; euhydrated ≤1.020) (3) were used.

Sensitivity and specificity were calculated to determine the accuracy of using Ucol and USG to determine hypohydrated (% BML ≥2.0) and euhydrated (%BML <2.0) states. Also, the accuracy of using Ucol to determine hypohydrated state through USG (USG >1.020) was also evaluated with hypohydrated value ≥5 and euhydrated value <5 as the criterion for Ucol.

To assess the potential carryover effect of each outdoor practice on the subsequent practice, paired samples t-tests assessed postpractice to the next day's prepractice values of Ucol, USG, and BM. A repeated-measures analysis of variance (ANOVA) was used to determine differences across practices. Specifically, Δ values = postoutdoor practice value − subsequent day's prepractice value. The seventh outdoor practice (Pr11) was omitted from the analysis of ΔUcol, ΔUSG, and ΔBM because it was the final preseason practice and thus there was not a subsequent practice. The difference in %BML across practices included all 11 outdoor practices. Pairwise comparisons examined differences by practice pairs (postpractice to next day's prepractice values).

Before the study, interrater and intrarater reliability was established by the single investigator who measured Ucol and USG for all subjects. Interrater reliability was established against an investigator with more thn 15 years of experience in taking the measures.

Effects sizes were calculated using Cohen's *d*, and effect sizes for carryover effects across practices were calculated using partial eta squared. Effect size categories were as follows: between 0.5 and 0.8, moderate and greater than 0.8, large (23). All *p* values were set at $p < 0.05$, and data were analyzed using SPSS statistical software (version 27, IBM Corporation, Armonk, NY).

Table 3

Urine color and urine-specific gravity diagnostic accuracy contingency table for body mass loss.*

	Number, condition	
	Hypohydrated (≥2% body mass loss)	Euhydrated (<2% body mass loss)
Urine color test outcome (score range)		
Positive/hypohydrated (5–8)	15 TP	82 FP
Negative/euhydrated (1–4)	10 FN	78 TN
Urine-specific gravity test outcome (score range)		
Positive/hypohydrated (>1.020)	15 TP	116 FP
Negative/euhydrated (≤1.020)	10 FN	44 TN

*TP = true positive; FP = false positive; FN = false negative; TN = true negative.

Table 4

Urine color diagnostic accuracy table for urine-specific gravity.*

	Number, condition	
Urine color test outcome (score range)	Hypohydrated (USG >1.020)	Euhydrated (USG ≤1.020)
Positive/hypohydrated (5–8)	89 TP	8 FP
Negative/euhydrated (1–4)	42 FN	46 TN

*TP = true positive; FP = false positive; FN = false negative; TN = true negative.

Copyright © 2023 National Strength and Conditioning Association. Unauthorized reproduction of this article is prohibited.

Downloaded from http://journals.lww.com/nsca-jscr by Ua4HyonDlxZ3azs6PvPeafPjdRjmqTz2+wOaEOfPRKBWqmu2eTwLZFZCxD413llzA9vtUjMVkNF8F/e29KG4jsPwh382C35tqtwKdeh21n/tkVAV08GvvOJ+sJCFOPfdY+hIYkKNuETHd/BLe1WCR3X4GJ6GTTKP1jwaHaj28= on 10/11/2023

Table 5
Carryover effects of one hot practice to the subsequent practice on urine color (Ucol).

	Urine color (Ucol)					
	Post Ucol	Pre Ucol	*p*	Effect size	95%	
Postpractice 2 to prepractice 3	5 ± 1	4 ± 1*	0.000	0.76	0.634	1.918
Postpractice 6 to prepractice 7 AM	5 ± 1	4 ± 1*	0.001	0.80	0.654	2.215
Postpractice 7 AM to prepractice 7 PM	5 ± 1	4 ± 1*	0.000	1.1	0.591	1.614
Postpractice 7 PM to prepractice 8	5 ± 1	4 ± 2	0.296	0.21	−0.372	1.172
Postpractice 8 to prepractice 9 AM	5 ± 1	5 ± 1	0.124	0.33	−0.154	1.198
Postpractice 10 to prepractice 11	5 ± 1	4 ± 1*	0.004	0.65	0.366	1.718

*Ucol—significant difference between postpractice and next day's prepractice measures.

Downloaded from http://journals.lww.com/nsca-jscr by Ua4HyonDIxZ3azs6PvPeafPjdRjmqTz2+wOaEOfPRKBWqmu2eTwLZFZCXD4138lzA9vtUjMVkNF8F/e29KG4jsPwh382C35tqtwKdeh21n/tkVAV08GvvOJ+sJCFOPfdY+hIYkKNuETHd/BLe1WCR3X4GJ6GTTKP1jwaHaj28= on 10/11/2023

Results

Sensitivity and Specificity of Using Percent of Body Mass Loss, Urine-Specific Gravity, and Urine Color to Determine Hydration

Using ROC analysis, the area under the curve (AUC) was not significant for Ucol or USG as a diagnostic tool to detect ≥2% BML (AUC = 0.55 and 0.49; p = 0.53 and 0.90, respectively). Sensitivity and specificity data for using UCol and USG to detect BML ≥2% are provided in Table 3. Sensitivity of detecting ≥2% BML was 60% for both. Specificity of detecting ≥2% BML using Ucol and USG to detect ≥2% BML was 48.8 and 44%, respectively.

Using ROC analysis, the AUC was significant for using Ucol to detect USG >1.020 (AUC = 0.85, p = 0.00) (Table 4). Sensitivity and specificity for using Ucol to determine USG (Table 3) were 82.5 and 67.9%, respectively.

Potential Carryover Effects of Measures of Hydration on the Subsequent Practice

Paired samples t-tests revealed postpractice to next day's prepractice differences in Ucol for practices 2–3 (t = 4.070; mean = 5 ± 1 and 4 ± 1; p = 0.000; effect size = 0.76; 95% CI = 0.634–1.918), 6–7 AM (t = 3.813; mean = 5 ± 1 and 4 ± 1; p = 0.001; effect size = 0.80; 95% CI = 0.654–2.215), 7 AM to 7 PM (t = 5.436; mean = 5 ± 1 and 4 ± 1; p = 0.001; effect size = 1.10; 95% CI = 1.110–2.473), and 10–11 (t = 3.187; mean = 5 ± 1 and 4 ± 1; p = 0.004; effect size = 0.65; 95% CI = 0.203–1.0863) (Table 5, Figure 2). There was also a significant difference in USG from practice 2 to practice 3 (t = 2.718; mean = 1.022 ± 0.007 and 1.018 ± 0.009; p = 0.011; effect size = 0.51; 95% CI = 0.001–0.007) (Table 6). There were no postpractice to the next day's prepractice differences in BM (Table 7). Effect sizes ranged from 0.12 to 0.24.

Carryover Effect Across Practices

Test-retest reliability measures, taken with 10 subjects over 2 days, revealed USG ICC = 0.99 and *SEM* = 0.00; Ucol ICC = 0.72 and *SEM* = 0.78 (Table 8). In looking at the carryover effect across practices (Table 9), there were no significant differences in ΔBM (F = 1.108; p = 0.359; effect size = 0.04). There were also no significant differences in ΔUSG (F = 0.753; p = 0.585; effect size = 0.02) across practices. This may be because of the relatively high standard deviation associated with USG. There were significant differences in ΔUcol (F = 2.580; p = 0.03; effect size = 0.08) where the differences from practice 2 to practice 3 (mean = 1.23 ± 1.69; 95% CI = 0.607–1.845) 6–7 AM (mean = 1.42 ± 1.61; 95% CI = 0.830–2.009), and 7 AM to 7 PM (mean = 1.52 ± 1.59; 95% CI = 0.933–2.099) were significantly greater than the difference from practice 8 to practice 9 AM (mean = 0.39 ± 1.38; 95% CI = −0.120 to 0.894).

Repeated-measures ANOVA revealed significant differences in %BML across practices (F = 9.399; p < 0.001; effect size = 0.725) (Table 10). The %BML of practices 7 PM (mean = 1.48 ± 0.75; 95% CI = 1.219–1.749) and 10 (mean = 1.32 ± 0.72; 95% CI = 1.070–1.577) was significantly greater than that of practice 1 or baseline (mean = 0.59 ± 0.72; 95% CI = 0.334–0.846), practice 2 (mean = 0.74 ± 0.66; 95% CI = 0.510–0.977), practice 6 (mean = 0.79 ± 0.80; 95% CI = 0.507–1.075), practice 7 AM (mean = 0.82 ± 0.61; 95% CI = 0.603–1.036), and practice 11 (mean = 0.71 ± 0.95; 95% CI = 0.366–1.048). The %BML of practice 8 (mean = 1.09 ± 0.87; 95% CI = 0.783–1.399) was significantly less than that of practice 7 PM (mean = 1.48 ± 0.75; 95% CI = 1.219–1.749) but not practice 10 (mean = 1.32 ± 0.72; 95% CI = 1.070–1.577).

Discussion

Hypohydration has been found to impair performance (7,21), increase fatigue (7,21), and reduce cardiovascular and

Table 6
Carryover effects of one hot practice to the subsequent practice on urine-specific gravity (USG).

	Urine-specific gravity (USG)					
	Post USG	Pre USG	*p*	ES	95%	
Postpractice 2 to prepractice 3	1.022 ± 0.007	1.018 ± 0.009*	0.002	0.51	0.010	0.007
Postpractice 6 to prepractice 7 AM	1.021 ± 0.007	1.020 ± 0.006	0.673	0.89	−0.321	0.498
Postpractice 7 AM to prepractice 7 PM	1.019 ± 0.007	1.018 ± 0.008	0.422	0.17	−0.238	0.568
Postpractice 7 PM to prepractice 8	1.022 ± 0.008	1.020 ± 0.009	0.306	0.21	−0.189	0.604
Postpractice 8 to prepractice 9 AM	1.024 ± 0.008	1.023 ± 0.006	0.668	0.09	−0.320	0.499
Postpractice 10 to prepractice 11	1.024 ± 0.007	1.023 ± 0.005	0.452	0.16	−0.248	0.557

*ES = effect size; Significant difference between post-practice and next day's pre-practice measures.

Copyright © 2023 National Strength and Conditioning Association. Unauthorized reproduction of this article is prohibited.

Downloaded from http://journals.lww.com/nsca-jscr by Ua4HyonDIxZ3azs6PvPeafPjdRjmqTz2+wOaEOfPRKBWqmu2eTwLZFZCxD413lIzA9vtUjMVkNF8F/e29KG4jsPwh382C35tqtwKdeh21n/tkVAV08GvvOJ+sJCFOPfdY+hIYkKNuETHd/BLe1WCR3X4GJ6GTTKP1jwaHaj28= on 10/11/2023

Table 7

Carryover effects of one hot practice to the subsequent practice on body mass (BM).

	Body mass					
	Postpractice BM (kg)	Prepractice BM (kg)	*p*	ES	95%	
Post-practice 2 to pre-practice 3	85.1 ± 18.8	85.6 ± 19.0	0.000	0.12	−2.720	5.238
Post-practice 6 to pre-practice 7 AM	84.7 ± 18.6	85.8 ± 19.1	0.259	0.24	−0.176	0.654
Postpractice 7 AM to prepractice 7 PM	85.1 ± 18.7	85.7 ± 19.0	0.452	0.16	−0.248	1.61
Postpractice 7 PM to prepractice 8	84.5 ± 18.7	85.3 ± 19.1	0.488	0.14	−0.255	0.533
Postpractice 8 to prepractice 9 AM	84.5 ± 18.6	85.1 ± 18.7	0.427	0.17	−0.245	0.578
Postpractice 10 to prepractice 11	84.6 ± 19.6	86.0 ± 19.9	0.314	0.22	−0.201	0.626

ES = effect size.

thermoregulatory response (2,11,12,18,19,25), which can exacerbate the risk of EHI, often before any clinical signs or symptoms appear. In addition, consecutive days of exercise in the heat can increase hypohydration and potentially the risk of EHI (22). Because adequate hydration becomes an important strategy in reducing EHI particularly in field settings in hot and humid environments, practical and reliable measures of hydration status in athletic settings would be important to identify.

Hypohydration is generally defined as 2–3% BML (4,13). This study sought to determine whether postexercise urinary hydration measures of Ucol and USG can be used to predict %BML in a field setting. It should be noted that although the practices included in our analysis had differing times between practices, this provides ecological validity. The results of our subjects do not suggest that either Ucol or USG alone can be used as a valid measure of hydration to detect %BML similar to that seen in our subjects because the sensitivity and specificity for both Ucol and USG were low (Table 2). The %BML of our high school athletes during preseason practices ranged from −0.9 to 3.4 with only 11.5% of %BML values greater than 2% across all outdoor practice days. Our results are in conflict with the results of McKenzie et al. (16) that determined that Ucol can accurately detect greater than 2% BML with a sensitivity of 88.9% and a specificity of 84.8%. Their study took place in a controlled laboratory environment that had their college-aged men exercise for 5 hours to force a 5% BML and produced BML ranging from 2 to 5%.

In a study examining military recruits, Wallace et al. (29) determined that the risk of EHI is determined by current day and previous day's WBGT. Repeated days of preseason football practices can have residual effects on heat strain and hydration if attention is not directed at replenishing fluids lost during exercise. Although chronic hydration status is difficult to assess in the field, being able to obtain hydration measures such as changes in BM, UC, and USG across days, can provide the clinician with an effective indication of changes in athletes' hydration over time. In reports of high school and collegiate football players, the baseline BM was greater than that of 5 (26) and 7 (10) consecutive days of football practice, respectively. In this study, the postpractice to next day's prepractice difference in BM was not statistically different, and the effect size was very small, which suggests no practical difference as well. This suggests that athletes were able to adequately rehydrate between practices. The prepractice values for UCol and USG provide support for this. It should be noted that the results might be different with a larger sample size or if the subjects were less hydrated. The subjects in this study were able to effectively rehydrate between practices.

In a study evaluating the effects of consecutive days of practice and competition on hydration status in youth soccer players, % BML, although mild, was greater on the second day of training (8). The %BML of the youth football players in the study by Stover et al. (26) (range = 0.04–1.3%) was less than that of the collegiate football players of Godek et al. (10) (range = 1.16–1.84%), but the environmental conditions of Stover et al. (26) were milder (WBGT = 15.5–23.2° C) than those of Godek et al. (10) (WBGT = 23.5–26.7° C). In looking at the effects of 2-a-day practices on hydration, Godek et al. (10) reported a greater %BML for afternoon practices in collegiate football players. It is noteworthy that the mean %BML for each of our outdoor practices ranged from 0.74 ± 0.66 to 1.48 ± 0.75%, with each day greater than the baseline %BML of 0.59 ± 0.72%. The practices with the greatest %BML were those after less than 24 hours between practices such as a 2-a-day practice or a morning practice after an evening practice. It should also be noted that equipment worn increased over the course of preseason practices in our study, which likely contributed to the increase in %BML. In a laboratory study evaluating consecutive days of exercise in the heat that mimicked preseason football practice, there was no difference in %BML on the second day of the study (22). However, fluid intake is more convenient in a laboratory study, and it is likely that subjects were able to more adequately rehydrate in the laboratory setting compared with an outdoor football practice in a field setting.

Urine-specific gravity values indicative of mild hypohydration with consecutive days of exercise in the heat have been reported in laboratory (22) and field studies (10,31). Our subjects began most subsequent practices after outdoor practices in a euhydrated state, which is evidenced by most USG of ≤1.020 except for the

Table 8

Intrarater reliability analysis of hydration measures.*

Variable	Day 1 Mean ± *SD*	Day 2 Mean ± *SD*	ICC (2,k)	SEM
Ucol	4.0 ± 1.3	3.7 ± 1.5	0.72	0.78
USG	1.0128 ± 0.0058	1.0128 ± 0.0059	0.99	0.00

**SEM* is expressed in units of the measure; ICC = intraclass correlation.

Table 9

Carryover effects on changes in body mass (ΔBM), urine-specific gravity (ΔUSG), and urine color (ΔUcol) across hot practices (mean ± *SD*; 95% confidence interval in parenthesis).*

	Δ body mass (BM) (kg)	Δ USG	ΔUcol
Postpractice 2 to prepractice 3	−0.52 ± 0.65 (−1.107 to −0.336)	0.003 ± 0.008 (0.001 to 0.006)	1.23 ± 1.69* (0.607 to 1.845)
Postpractice 6 to prepractice 7 AM	−1.17 ± 1.16 (−1.412 to −0.346)	0.000 ± 0.008 (−0.003 to 0.003)	1.42 ± 1.61* (0.830 to 2.009)
Postpractice 7 AM to prepractice 7 PM	−0.69 ± 0.71 (−0.898 to −0.138)	0.002 ± 0.007 (−0.001 to 0.004)	1.52 ± 1.59* (0.933 to 2.099)
Postpractice 7 PM to prepractice 8	−0.85 ± 1.52 (−1.267 to 0.067)	0.002 ± 0.009 (−0.001 to 0.006)	0.58 ± 1.88 (−0.107 to 1.269)
Postpractice 8 to prepractice 9 AM	−0.56 ± 0.91 (−0.857 to 0.157)	0.001 ± 0.008 (−0.002 to 0.004)	0.39 ± 1.38 (−0.120 to 0.894)
Postpractice 10 to prepractice 11	−1.36 ± 0.78 (−1.347 to −0.459)	0.001 ± 0.008 (−0.002 to 0.003)	0.84 ± 1.49 (0.294 to 1.384)

*ΔUcol—significantly greater than postpractice 8 to prepractice 9 AM

Copyright © 2023 National Strength and Conditioning Association. Unauthorized reproduction of this article is prohibited.

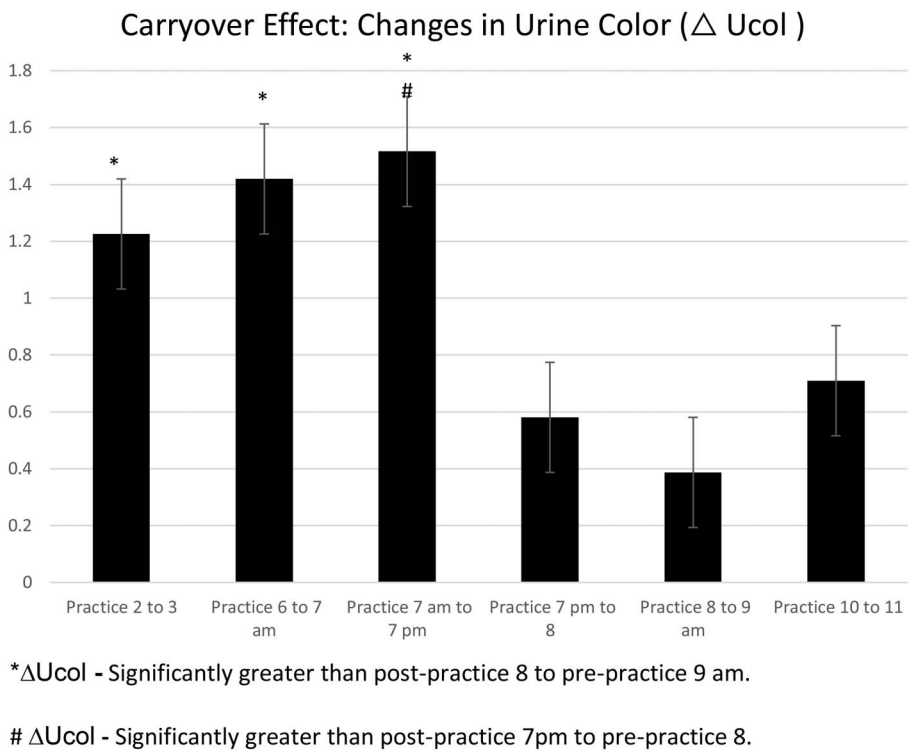

Figure 2. Difference in urine color between postoutdoor practice and next prepractice measure.

prepractice USG values after the last 2 practices in the heat, which were >1.020, indicative of hypohydration. Our study provides evidence of good sensitivity and specificity for using Ucol to determine USG. However, neither USG nor Ucol for our population was able to be used to detect >2% BML, which is indicative of impaired thermoregulation. This is not unexpected as our subjects did not experience that level of %BML. Our results provide evidence of a carryover effect in Ucol between practices. The postpractice to prepractice differences were greater after the first outdoor practice (practice 2) and for practices with less than 24 hours between practices. For most practices, the effect sizes were moderate, suggesting some practical significance. These findings support the notion that fluid recommendations should emphasize the need to replenish fluids after each day and especially after 2-a-day practices; otherwise, athletes' hydration may suffer over subsequent days and practices.

Table 10

Carryover effects on changes in body mass loss (BML) across hot practices.

	Body mass loss (BML) (%)		
		95% confidence interval	
	Mean ± *SD*	Lower bound	Upper bound
Baseline	0.59 ± 0.72*	0.334	0.846
Postpractice 2	0.74 ± 0.66*	0.510	0.977
Postpractice 6	0.79 ± 0.80*	0.507	1.075
Postpractice 7 AM	0.82 ± 0.61*	0.603	1.036
Postpractice 7 PM	1.48 ± 0.75†	1.219	1.749
Postpractice 8	1.09 ± 0.87‡	0.783	1.399
Postpractice 10	1.32 ± 0.72	1.070	1.577
Postpractice 11	0.71 ± 0.95*	0.366	1.048

*%BML—significantly less than practice 7 PM and 10.
†%BML—significantly greater than all other practices except practice 10.
‡%BML—significantly different from practice 7 PM, but not practice 10.

Practical Applications

Preseason American football practices in a hot environment incur a significant risk of EHI. As adequate hydration is important to reduce that risk, practical and valid assessments of hydration status should be determined. Because no single measure of hydration status is without limitations, multiple assessments may help reduce misclassification of hydration. The high school athletes in our study began most practices at the upper end of acceptable hydration evidenced by USG of ≤1.020 and UCol of <5. The Ucol values along with the minimal %BML suggest that our subjects were likely not at significant risk of EHI. Consecutive practices can increase the risk of hypohydration, a risk for EHI. Based on our % BML, there was a carryover effect from one hot practice to the next practice after 2-a-day practices, although the % BML did not reach a critical level. The changes in Ucol and USG represent a cumulative effect in the latter days. As such, it is prudent to ensure that athletes are adequately hydrating, especially after 2-a-day practices. The athletes in this study were able to effectively replenish fluids lost with some evidence of reduced effectiveness, suggesting that the effects of previous days' practices were cumulative. These results provide credence that athletes must purposefully replenish fluids during and after every practice to maintain adequate euhydration, particularly with multiple practices in hot, humid environments.

Copyright © 2023 National Strength and Conditioning Association. Unauthorized reproduction of this article is prohibited.

Downloaded from http://journals.lww.com/nsca-jscr by Ua4HyonDIxZ3azs6PvPeafPjdRjmqTz2+wOaEOfPRKBWqmu2eTwLZFZCxD413lIzA9vtUjMVkNF8F/e29KG4jsPwh382C35tqtwKdeh21n/tkVAV08GvOJ+sJCFOPfdY+hIYkKNuETHd/BLe1WCR3X4GJ6GTTKP1jwaHaj28= on 10/11/2023

Acknowledgments

The authors thank all subjects in the study and all students, coaches, and staff who assisted with this project. This study was funded by the National Athletic Trainers' Association Research & Education Foundation Grant (General Grant #1819GGP02). None of the authors or the National Strength and Conditioning Association has any commercial or proprietary interest in any instrument that was used in this article.

Downloaded from http://journals.lww.com/nsca-jscr by Ua4HyonDIxZ3azs6PvPeafPjdRjmqTz2+wOaEOfPRKBWqmu2eTwLZFZCxD413lIzA9vtUjMVkNF8F/e29KG4jsPwh382C35tqtwKdeh21n/tkVAV08GvvOJ+sJCFOPfdY+hIYkKNuETHd/BLe1WCR3X4GJ6GTTKP1jwaHaj28= on 10/11/2023

References

1. Adams WM, Vandermark LW, Belval LN, Casa DJ. The utility of thirst as a measure of hydration status following exercise-induced dehydration. *Nutrients* 11: 2689, 2019.
2. Armstrong LE, Maresh CM, Castellani JW, et al. Urinary indices of hydration status. *Int J Sport Nutr* 4: 265–279, 1994.
3. Barley OR, Chapman DW, Abbiss CR. Reviewing the current methods of assessing hydration in athletes. *J Int Soc Sports Nutr* 17: 1–13, 2020.
4. Casa DJ, DeMartini JK, Bergeron MF, et al. National Athletic Trainers' Association position statemens: Exertional heat illnesses. *J Athl Train* 50: 986–1000, 2015.
5. DaSilva RP, Mundel T, Natali AJ, et al. Pre-game hydration status, sweat loss, and fluid intake in elite Brazillian male soccer players during competition. *J Sports Sci* 30: 37–42, 2012.
6. DeMartini-Nolan JK, Martschinske JL, Casa DJ, et al. Examining the influence of exercise intensity and hydration on gastrointestinal temperature in collegiate football players. *J Strength Cond Res* 32: 2888–2896, 2018.
7. Deshayes TA, Jeker D, Goulet EDB. Impact of pre-exercise hypohydration on aerobic exercise performance, peak oxygen consumption and oxygen consumption at lactate threshold: A systematic review with meta-analysis. *Sports Med* 50: 581–596, 2020.
8. Ersoy N, Ersoy G, Kutlu M. Assessment of hydration status of elite young male soccer players with different methods and new approach method of substitute urine strip. *J Int Soc Sports Nutr* 13: 1–6, 2016.
9. Gamage PJ, Fortington LV, Finch CF. Epidemiology of exertional heat illnesses in organised sports: A systematic review. *J Sci Med Sport* 23: 701–709, 2020.
10. Godek SF, Godek JJ, Bartolozzi AR. Hydration status in college football players during consecutive days of twice-a-day preseason practices. *Am J Sports Med* 33: 843–851, 2005.
11. Gonzalez-Alonso J, Mora-Rodriguez R, Below PR, Coyle EF. Dehydration reduces cardiac output and increases systemic and cutaneous vascular resistance during exercise. *J Appl Physiol* 79: 1487–1496, 1995.
12. González-Alonso J, Mora-Rodríguez R, Below PR, Coyle EF. Dehydration markedly impairs cardiovascular function in hyperthermic endurance athletes during exercise. *J Appl Physiol* 82: 1229–1236, 1997.
13. Greenleaf JE. Problem: Thirst, drinking behavior, and involuntary dehydration. *Med Sci Sports Exerc* 24: 645–656, 1992.
14. Kerr ZY, Register-Mihalik JK, Pryor RR, et al. The association between mandated preseason heat acclimatization guidelines and exertional heat illness during preseason high school American football practices. *Environ Health Perspect* 127: 47003-1–47003-6, 2019.
15. McDermott BP, Anderson SA, Armstrong LE, et al. National Athletic Trainers' Association position statement: Fluid replacement for the physically active. *J Athl Train* 52: 877–895, 2017.
16. McKenzie AL, Munoz CX, Armstrong LE. Accuracy of urine color to detect equal to or greater than 2% body mass loss in men. *J Athl Train* 50: 1306–1309, 2015.
17. Meyer F, Volterman KA, Timmons BW, Wilk B. Fluid balance and dehydration in the young athlete:assessment considerations and effects on health and performance. *Am J Lifestyle Med* 6: 489–501, 2012.
18. Montain S, Sawka M, Latzka WA, Valeri C. Thermal and cardiovascular strain from hypohydration: Influence of exercise intensity. *Int J Sports Med* 19: 87–91, 1998.
19. Montain SJ, Coyle EF. Influence of graded dehydration on hyperthermia and cardiovascular drift during exercise. *J Appl Physiol* 73: 1340–1350, 1992.
20. Munoz CX, Johnson EC, DeMartini JK, et al. Assessment of hydration biomarkers including salivary osmolality during passive and active hydration. *Eur J Clin Nutr* 67: 1257–1263, 2013.
21. Périard JD, Eijsvogels TMH, Daanen HAM. Exercise under heat stress: Thermoregulation, hydration, performance implications, and mitigation strategies. *Physiol Rev* 101: 1873–1979, 2021.
22. Pryor RR, Pryor JL, Vandermark LW, et al. Exacerbated heat strain during consecutive days of repeated exercise sessions in heat. *J Sci Med Sport* 22: 1084–1089, 2019.
23. Sawilowsky SS. New effect size rules of thumb. *J Mod Appl Stat Meth* 8: 597–599, 2009.
24. Sawka MN, Burke LM, Eichner ER, Maughan RJ, Montain SJ, Stachenfeld NS. American college of sports medicine position stand: Exercise and fluid replacement. *Med Sci Sports Exer* 39: 377–390, 2007.
25. Sawka MN, Francesconi RP, Young AJ, Pandolf KB. Influence of hydration level and body fluids on exercise performance in the heat. *JAMA* 252: 1165–1169, 1984.
26. Stover EA, Zachwieja J, Stofan J, Murray R, Horswill CA. Consistently high urine specific gravity in adolescent American football players and the impact of an acute drinking strategy. *Int J Sports Med* 27: 330–335, 2006.
27. Tripp BL, Eberman LE, Smith MS. Exertional heat illnesses and environmental conditions during high school football practices. *Am J Sports Med* 43: 2490–2495, 2015.
28. Volpe SL, Poule KA, Bland EG. Estimation of prepractice hydration status of national collegiate athletic association Division I athletes. *J Athl Train* 44: 624–629, 2009.
29. Wallace RF, Kriebel D, Punnett L, et al. The effects of continuous hot weather training on RIsk of exertional heat illness. *Med Sci Sports Exer* 37: 84–90, 2005.
30. Yard EE, Gilchrist J, Haileyesus T, et al. Heat illness among high school athletes—United States, 2005-2009. *J Saf Res* 41: 471–474, 2010.
31. Yeargin SW, Casa DJ, Judelson DA, et al. Thermoregulatory responses and hydration practices in heat-acclimatized adolescents during preseason high school football. *J Athl Train* 45: 136–146, 2010.
32. Yeargin SW, Dickinson JJ, Emerson DM, et al. Exertional heat illness risk factors and physiological responses of youth football players. *J Sport Health Sci* 10: 91–98, 2021.
33. Yeargin SW, Dompier TP, Casa DJ, Hirschhorn RM, Kerr ZY. Epidemiology of exertional heat illnesses in national collegiate athletic association athletes during the 2009-2010 through 2014-2015 academic years. *J Athl Train* 54: 55–63, 2019.
34. Yeargin SW, Kerr ZY, Casa DJ, et al. Epidemiology of exertional heat illnesses in youth, high school, and college football. *Med Sci Sports Exerc* 48: 1523–1529, 2016.

Copyright © 2023 National Strength and Conditioning Association. Unauthorized reproduction of this article is prohibited.